EXP.NO: 6	Write a Servlet to demonstrate session tracking using HttpSession.

AIM:

To demonstrate session tracking using HttpSession in a Servlet-based web application. The exercise involves implementing a simple login system where a user can log in with their username and password, and their session will be tracked throughout their interaction with the web application.

ALGORITHM:

1. Input:

- The user will enter their **username** and **password** in a **login form**.
- The servlet will validate the credentials and create an **HTTP session** for the user if authentication is successful.

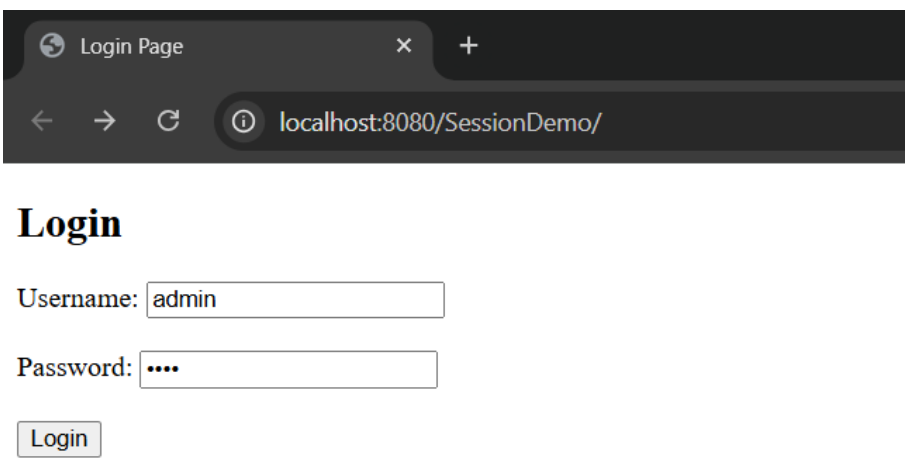
2. Steps for Implementing Session Tracking with HttpSession:

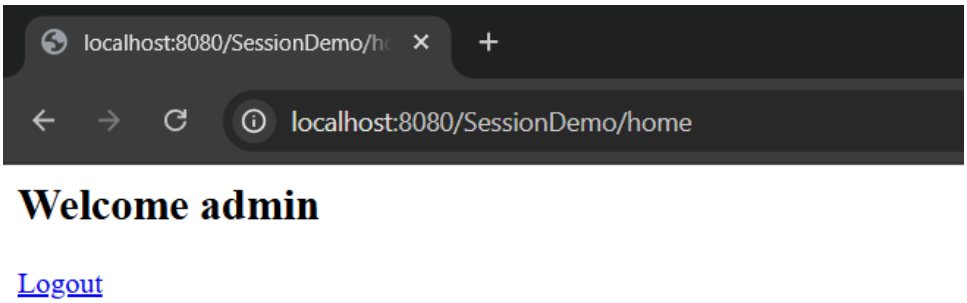
1. **Create the Login Form (HTML):**
    a. Create an HTML form with two input fields for **username** and **password**.
    b. Provide a **submit button** to send the data to the servlet for authentication.
2. **Create the Login Servlet:**
    a. Write a Java servlet class to handle **login** requests.
    b. The servlet will:
        i. Retrieve the **username** and **password** from the form.
        ii. Authenticate the user by comparing the credentials to a stored value (e.g., hardcoded or from a database).
        iii. Create a new **HTTP session** for the user if the credentials are valid.
        iv. Set session attributes such as the user's **username** to track the session.
3. **Validate User Credentials:**
    a. In the servlet, check if the **username** and **password** match the expected values.
    b. If valid, create a session and store the **username** in the session.
    c. If invalid, redirect the user back to the login page with an error message.
4. **Create and Track User Session:**
    a. Upon successful login, use the HttpSession object to store the session data.
    b. Use the session.setAttribute("username", username) method to save the **username**.

c. Optionally, set session timeout using session.setMaxInactiveInterval(time_in_seconds) to manage session expiration.

5. **Redirect to a Protected Page:**
   a. After a successful login, the user will be redirected to a **protected page** (e.g., a dashboard or home page).
   b. The protected page should check if the user is authenticated by checking the **session attribute**.
6. **Logout and Session Invalidation:**
   a. Provide a **logout feature** that invalidates the session using session.invalidate() when the user logs out.
   b. After logout, redirect the user back to the login page.
7. **Session Expiration:**
   a. If the session expires or the user manually logs out, the session will be invalidated, and the user will be redirected to the login page.

Output:





HTML File (index.html) :

```
<!DOCTYPE html>
<html>
<head>
    <title>Login Page</title>
</head>
<body>
<h2>Login</h2>
<form action="LoginServlet" method="post">
    Username: <input type="text" name="username"><br><br>
    Password: <input type="password" name="password"><br><br>
    <input type="submit" value="Login">
</form>
</body>
</html>
```

Java File (LoginServlet.java):

```
package com.example;
```

```
import java.io.IOException;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.WebServlet;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        String username = request.getParameter("username");
        String password = request.getParameter("password");
        if (username.equals("admin") && password.equals("1234")) {

            HttpSession session = request.getSession();
            session.setAttribute("username", username);
            session.setMaxInactiveInterval(300);
            response.sendRedirect("home");
        } else {
            response.getWriter().println("<h3>Invalid login! Try again.</h3>");
        }
    }
}
```

Java File (HomeServlet.java):

```
package com.example;
import java.io.IOException;
import java.io.PrintWriter;
```

```
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.WebServlet;

@WebServlet("/home")
public class HomeServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        HttpSession session = request.getSession(false);

        if (session != null && session.getAttribute("username") != null) {
            String user = (String) session.getAttribute("username");
            out.println("<h2>Welcome " + user + "</h2>");
            out.println("<a href='logout'>Logout</a>");
        } else {
            response.sendRedirect("index.html");
        }
    }
}
```

Java File (LogoutServlet.java):

```
package com.example;

import java.io.IOException;


import jakarta.servlet.*;

import jakarta.servlet.http.*;

import jakarta.servlet.annotation.WebServlet;


@WebServlet("/logout")

public class LogoutServlet extends HttpServlet {


    protected void doGet(HttpServletRequest request, HttpServletResponse response)

            throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if (session != null) {

            session.invalidate(); // destroy session

        }

        response.sendRedirect("index.html");

    }

}
```

RESULT:

A simple login system is created where the user's session is tracked by a servlet to demonstrate session tracking using HttpSession.